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Nana Porter-Honicky

Computational + Experimental Biomechanics

I use machine learning to study the mechanics of human joints in clinical populations



Education

Ph.D. in Mechanical Engineering

MechE Dept., University of Michigan 🌐

Ann Arbor, MI
08/23–**present**

- Advised by [Elliott Rouse](#) 🌐 (MechE/Robotics)
- Thesis topic: **Estimating Joint Impedance in Clinical Populations using Machine Learning**

M.Sc. in Movement Science

Sch. of Kinesiology, University of Michigan 🌐

Ann Arbor, MI
08/23–**present**

B.Sc. in Mechanical Engineering

UC Berkeley 🌐

Berkeley, CA
08/21–05/23

Research Experience

Directed Study & Research

MechE Dept., University of Michigan 🌐

Ann Arbor, MI
08/23–**present**

- Advised by [Dr. Elliott Rouse](#) 🌐 (MechE/Robotics)
- Develop machine learning models to forecast joint kinematics and kinetics
- Collect and analyze data from human subjects with musculoskeletal impairments
- Collaborate with clinicians to develop an exo-skeleton to assist patients with nerve injury

Directed Study & Research

Physics Division, Lawrence Berkeley National Laboratory 🌐

Berkeley, CA
08/21–05/23

- Advised by [Dr. Peter Sorensen](#) 🌐 (Physics)
- Design of mold for the fabrication of scintillating PTFE material
- Design of dark box components for use in testing scintillating PTFE
- Design of vertical circuit to modify the electromagnetic field in the detector

Work Experience

Berkeley Lab Undergraduate Research Intern

Lawrence Berkeley National Laboratory 🌐

Berkeley, CA
05/21–08/21

- Designed and fabricated components of a heatsink for Kapton cables in the CrystaLiZe dark matter detector

Electronics Lab Design and Implementation Intern

Tarana Wireless 🌐

Santa Clara, CA
05/19–08/19

- Fulfilled IT requests for RF laboratory installation and maintenance
- Designed layout for new RF lab, including power, grounding, and network connections

Technical Skills

OpenSim MATLAB Simulink PyTorch CUDA Git

Docker Python C++ Solidworks Onshape Vicon Nexus Delsys EMGworks $\LaTeX$ WandB

optimization linear algebra probability theory signal processing control theory

Conference Proceedings

N. Porter-Honicky and E. Rouse, “Joint impedance estimation using multi-task deep learning for gait kinematic and kinetic forecasting,” in *BioRob*, 2026. In submission

Invited Talks

Fabrication of ZnO-PTFE material for use in LUX-ZEPLIN dark matter detection experiment Los Angeles, CA
to flag lead-210 alpha decays through scintillation
UCLA, *UCLA National McNair Conference* ☞ 07/22

Poster Presentations

Forecasting Gait Kinetics and Kinematics for Biological Joint Impedance Estimation Using Machine Learning Ma-Chicago, IL
University of Chicago, *Midwest Machine Learning Symposium* ☞ 06/25

★ **Best Poster Award**

Forecasting Gait Kinetics and Kinematics for Biological Joint Impedance Estimation Using Machine Learning Ma-Chicago, IL
RehabWeek, 2025 International Consortium of Rehabilitation Robotics ☞ 05/25

Using Kinetic and Kinematic Prediction Models to Characterize Ankle Impedance Pensicola, FL
IHMC, Dynamic Walking Conference ☞ 05/24

Awards and Honors

Best Poster Award Chicago, IL
University of Chicago, *Midwest Machine Learning Symposium* ☞ 05/25

Rackham Merit Fellowship Ann Arbor, MI
University of Michigan 05/23-present

- Covers two full years of tuition, stipend, and health/dental insurance.

McNair Scholarship Berkeley, CA
UC Berkeley, *McNair Scholars Program* ☞ 2021 cohort

- Received funding for my work at LBNL, gave a presentation at the McNair 2022 Symposium at UCLA, and published in the 2022 Berkeley McNair Research Journal

ME Scholar Berkeley, CA
UC Berkeley 2021 cohort

- Hand-picked group of incoming MechE students at UC Berkeley chosen by the MechE counselor for a weekly seminar involving a stipend, research presentations by faculty, and career/academic planning

NSF S-STEM Scholarship Aptos, CA
Cabrillo College 2018 cohort

- SPECS Scholarship Program supports academically talented, financially disadvantaged students from groups that are traditionally underrepresented in engineering professions: minorities, women, and first-generation students

Teaching and Mentorship

Career Day Speaker Ann Arbor, MI
Washtenaw Community College 02/26

- Gave a presentation to students about my research and career path, what it means to be a Mechanical Engineer, and led a Lego building competition to illustrate the engineering design process

Graduate Student Instructor Ann Arbor, MI
University of Michigan 08/25–12/25

- Taught a weekly lab section of 30 students where we covered CAD, modeling dynamic systems, circuits, controls, etc.
- Held weekly office hours to assist students with homework and projects
- Wrote and graded midterm and final exams
- Design and Manufacturing II(*ME 350* ☞)

Planner and Instructor Ann Arbor, MI
Engineering Day with the Girl Scouts 04/25

- Taught Junior and Brownie Girl Scouts about Mechanical Engineering with forces and motion

Planner and Instructor Ann Arbor, MI
Detroit Area Pre-College Engineering Program (DAPCEP) 02/25-03/25

- Taught High School and Middle School students engineering fundamentals such as forces and motion, mechanical design, fluid mechanics, programming etc.

Instructor Ann Arbor, MI
Detroit Area Pre-College Engineering Program (DAPCEP) 02/24-03/24

- Taught High School and Middle School students engineering fundamentals such as forces and motion, mechanical design, fluid mechanics, programming etc.

Instructor Detroit, MI
First Robotics at the International Conference on Intelligent Robots and Systems (IROS) 10/23

- Taught High School Students about programming principles for robotics

Service

Planner for the Disability Visibility in Engineering Symposium Ann Arbor, MI
University of Michigan 04/26

- A student-driven symposium that creates a space for the engineering community to discuss and learn about ways to build more accessible engineering environments, research methods, and technologies.

Coordinator for Tech + Culture Reading Group Ann Arbor, MI
University of Michigan 08/25–present

- A biweekly student led reading group that explores the roles of technology in society, and its impact on things like disability, labor, fiction and more.

Outreach Chair Ann Arbor, MI
University of Michigan, Mechanical Engineering Graduate Council (MEGC) ∞ 05/24–present

PR Chair Ann Arbor, MI
University of Michigan, Women and Gender Minorities in Mechanical Engineering (WaGMiME) ∞ 05/24–present

Class Projects

CycloStride: Design and Development of a Prosthetic Knee Optimized for Level and Ramp Walking Ann Arbor, MI

Locomotor Mechanics and Design / Control of Wearable Robotic Systems Fall 2025

- This project aimed to design and model the thermal properties and controls for a powered prosthetic knee with a cycloid transmission for use on level ground and sloped walking.

Dead reckoning on the Össur Power Knee Ann Arbor, MI
Quantifying Human Motion Through Wearable Sensors Winter 2025

- This study explores the feasibility of using the limited sensor suite of the Ossur Power Knee to estimate gait speed and incline in a pilot study using dead reckoning.

Effects of Powered Ankle Prostheses on Lower Limb Kinetics and Metabolics: A Literature Review Ann Arbor, MI

Clinical Gait Analysis Fall 2024

- A semester long literature review on the impact of powered ankle prostheses on sloped walking in unilateral transtibial amputees.

Wrist-guard Pedal for Hand Cycle Berkeley, CA
Augmenting Human Dexterity Spring 2023

- An adapted off-the-shelf wrist guard that can attach to a hand-cycle with a magnetic mechanism. Designed to alleviate wrist pain for hand-cycle user. Patent submitted for the magnetic attachment mechanism.

Sous-Gardener II*Design of Microprocessor-Based Mechanical Systems*

Berkeley, CA

Spring 2023

- An upgrade to the Sous-Gardener to also track light levels and actuate a pump or a grow-light when the soil is too dry or the plant isn't getting enough light.
- A new GUI for the Sous-Gardener that allows you to manually toggle the grow-light and pump.

Beer Pour Machine*Mechatronics Design*

Berkeley, CA

Fall 2022

- Device that pours your beer for you to get the perfect head.
- Using a linkage designed to make the optimal pour from a bottle with another actuator to tilt your glass just right so you don't get too much foam.

Sous-Gardener I*Internet of Things*

Berkeley, CA

Spring 2022

- A small flowerpot with an embedded moisture sensor and NFC tag. When you tap the flower pot with your phone it takes you to a webpage with live readouts from the moisture sensor and tells you if you need to water your plant.